RUSTAM TIMALSINA

Full-Stack & Systems Engineer | Python (FastAPI), React, Node.js, Laravel | BSc CSIT

Kathmandu, Nepal • (+977) 9848915588 • rustamtimalsina179@gmail.com • Portfolio • LinkedIn • GitHub

TECHNICAL SKILLS

- **Languages:** Python, JavaScript (ES6+), TypeScript, PHP, C, C++, SQL, HTML5/CSS3
- **Backend & APIs:** FastAPI, Node.js, Express.js, Laravel, RESTful APIs, WebSockets (Socket.io), JWT, RBAC
- **Frontend:** React.js, Vite, Tailwind CSS, Component Architecture, State Management, Three.js
- **Databases:** PostgreSQL, MySQL, MongoDB Atlas, SQLite, Schema Normalization, Relational Indexing
- **DevOps & Tools:** Git, GitHub, Linux, Postman, Vercel, Render, Scikit-Learn, Information Retrieval (BM25)

WORK EXPERIENCE

Software Developer Intern | Bytherix *(Remote • Sep 2026 – Present)*

- **Engineered ClientHub:** a centralized milestone management portal using Laravel (PHP) and MySQL.
- **Architected a finite-state machine** to govern deliverable lifecycles across review, approved, and paid milestones with immutable audit trails.
- **Implemented Role-Based Access Control (RBAC)** to strictly isolate freelancer and enterprise client permissions with CSRF token protection.
- **Designed normalized schemas** with foreign keys and indexing, reducing query latency and eliminating orphaned records.

Full-Stack Developer Intern | Syntecxhub *(Remote • Sep 2026 – Present)*

- **Architected Lanes:** a MERN task management platform featuring drag-and-drop boards, JWT auth, and API rate limiting.
- **Engineered Production Services:** FastAPI backends with SQLite data isolation, OpenWeatherMap proxy integration, and client caching.
- **Enforced Strict Security:** with Postman end-to-end verification, input sanitization, and CORS headers across all endpoints.

FEATURED PROJECTS

NexusAI Studio — Multi-Persona AI Workspace | *React, TypeScript, Node.js, Gemini API, SSE* **[Live Demo ↗] [GitHub]**

- **Low-Latency Streaming:** a real-time Server-Sent Events (SSE) streaming pipeline delivering <50ms Time to First Token (TTFT).
- **Integrated LLM Engine:** Gemini API with custom persona prompts, conversational memory buffers, and offline simulations.

Nightdesk — Real-Time WebSockets Chat Platform | *React, Node.js, Socket.io, Three.js, MongoDB* **[Live Demo ↗] [GitHub]**

- **Full-Stack WebSockets:** a MERN messaging app with JWT-authenticated WebSocket handshakes, 1-on-1 DMs, and public rooms.
- **Rich Features & 3D UI:** typing indicators, online presence, message quoting, and an interactive 3D scene using Three.js.

NestHub — Property Platform & FastAPI REST Backend | *Python (FastAPI), React, SQLite, JWT, Swagger* **[Live Demo ↗] [GitHub]**

- **Scalable Backend:** production REST APIs with interactive OpenAPI docs, Pydantic validation, and role-based authorization.
- **Relational Schemas:** property filtering, favorites management, and normalized queries with sub-50ms execution.

Uriel — Domain-Specific Search Engine | *Python, FastAPI, Scikit-Learn, BM25, TF-IDF* **[GitHub]**

- **IR Architecture:** an Information Retrieval search engine with custom inverted indexing and BM25 ranking algorithms.

EDUCATION & CREDENTIALS

BSc in Computer Science & Information Technology (CSIT) | Amrit Science Campus (Tribhuvan University) *(Kathmandu • 2022 – 2027)*

- **Cumulative GPA:** 3.26 / 4.0 (Completed 6th Semester) • **Key Coursework:** Data Structures & Algorithms, Artificial Intelligence, DBMS, NLP, Linear Algebra
- **Software Fellowship:** Selected for competitive fellowship at LOCUS 2026, IOE Pulchowk Campus.
- **Cybersecurity Job Simulation:** Mastercard (via Forage) — credentialed security and endpoint protection simulation.